3.7 Nitrous oxide (N2O) analysis
An FTIR analyser, an NDIR (non -dispersive infrared) analyser, laser infrared analyser or NDUV analyser may be used in accordance with the instrument supplier's instructions.
Note:
According to sections 5.4.2 and 5.4.3 of the NTC 2008, other systems or analysers may, subject to the approval of the Administration, be accepted if they yield equivalent results to that of the equipment referenced.
Appendix IV Calibration of the analytical and measurement instruments
Requirements of table 1 or table 3 regarding fuel measurement device apply separately to both liquid fuel meter and gas fuel meter.
In addition to the calibration procedures of appendix IV of the NTC 2008, relevant parts of ISO 8178 -1 for calibration of CH 4 and/or N2O measurement instruments should be applied.
FID should be calibrated with NMC bypassed with NMC

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SOURCE: MEPC.402(83).pdf
In addition to the calibration procedures of appendix IV of the NTC 2008, relevant parts of ISO 8178 -1 for calibration of CH 4 and/or N2O measurement instruments should be applied.
FID should be calibrated with NMC bypassed with NMC
efficiencies Ee and Em determined separately.
Section 2
"C2H6 and purified synthetic air" is added to 2.2.1.5.
A new sub-paragraph "2.2.1.6 N2O and purified nitrogen" is added after the existing 2.2.1.5.
Section 5
N2O is added to the list of analysers in the first sentence of 5.4.2.
8.4 For measurement of N2O, a new section 8.4 applies
8.4.1 The interference should be checked prior to first use of an analyser and after major servicing or updating of software.
8.4.2 In those cases where the analyser applies compensation algorithms which use as inputs the concentrations of other measured gases those measurements should be undertaken concurrently with this verification check.
8.4.3 For NDIR - The potential for cross interferences effects of CO, CO2, H2O, CH4 and SO2 as applicable should be checked.
8.4.3.1 Apart from H 2O the interference species are dependent on the selected N2O absorption band used by the device which should

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SOURCE: MEPC.402(83).pdf
CO2, H2O, CH4 and SO2 as applicable should be checked.
8.4.3.1 Apart from H 2O the interference species are dependent on the selected N2O absorption band used by the device which should

be known. From that knowledge good engineering judgement should be used to determine the interference gases to be used based on those which may be expected to be present in the exhaust gases to be measured.

8.4.4 For FTIR or Laser Infrared The inference gases depend on the selected N2O absorption band used by the device which should be known. Based on that knowledge good engineering judgement should be used to determine the interference gases to be used.
8.4.5 The combined effect of the interference gases should not be more 1.0 µmol/mol.
8.4.6 Verification procedure:
.1 The concentrations of the interference span gases as identified from 8.4.3 or 8.4.4 should be at least as high as the maximum values to be encountered in service. Those

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SOURCE: MEPC.402(83).pdf
more 1.0 µmol/mol.
8.4.6 Verification procedure:
.1 The concentrations of the interference span gases as identified from 8.4.3 or 8.4.4 should be at least as high as the maximum values to be encountered in service. Those
interference gases may be presented in the form of a multi - component span gas.
.2 The N2O analyser is started, operated, zeroed and spanned as in service.
.3 Humified interference test gas should be fed into the analyser. That test gas should be generated by bubbling the multi-component span gas through distilled H 2O in a sealed container. If the sample is not treated by a dryer the container temperature should be controlled to generate a H 2O concentration at least as high as the expected maximum when in service. If the sample is treated by a dryer the container temperature should be controlled to generate a H2O concentration at least as high as the expected maximum based on the dryer outlet temperature when in service.
.4 The water mole fraction of the test gas should be determined from measurements taken as close as possible to the analyser inlet. Those measurements may be dew point and absolute pressure.

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.4 The water mole fraction of the test gas should be determined from measurements taken as close as possible to the analyser inlet. Those measurements may be dew point and absolute pressure.

.5 Condensation in the piping leading from the container generating the humidified test gas to the analyser should be minimized by maintaining an adequate minimum temperature.
.6 Following stabilization, the analyser output should be recorded for 30 s. The arithmetic mean response over that period should be compared with the limit in 8.4.5.
8.4.6.1 As an alternative to the multi -component span gas in 8.4.6 individual span gases may be run separately.
Where an interference gas concentration is higher than that to be measured in service the determined interference value should be scaled down by the ratio of in -service maximum / span concentration.
Where the H 2O concentration is below that to be measured in service, but not below 0.025 mol/mol H2O content, the determined

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interference may be scaled up by the ratio of maximum in-service value / the value used.
The sum of the individual interferences should be compared with the limit in 8.4.5.
8.4.7 An interference verification check report documenting the procedure as followed, including the rational for the interference gases used and their concentrations, and the outcomes of that procedure is to be prepared and should be available as may be required.
8.4.8 Alternative approaches to the verification of N 2O analyser interference may be acceptable. Where so used the justification for the approach taken should be included in the report as required by 8.4.7.
Irrespective of the procedure followed the limit given by 8.4.5 remains applicable.
8.5 Efficiency of the non-methane cutter (NMC)
The NMC is used for the removal of the non -methane hydrocarbons from the sample gas by oxidizing all hydrocarbons

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remains applicable.
8.5 Efficiency of the non-methane cutter (NMC)
The NMC is used for the removal of the non -methane hydrocarbons from the sample gas by oxidizing all hydrocarbons
except CH4. Ideally, the conversion rate for CH4 is 0% and for the

other hydrocarbons, as represented by ethane, is 100%. Since the performance of NMC can deteriorate rapidly and without warning if operated outside certain ranges of gas concentrations and temperature ranges, the efficiency of the NMC sh ould be checked as part of the pretest verification procedures under 6.1 and again on completion of the measurement exercise (at the time of rechecking the analysers in accordance with 5.9.9 of NTC 2008) with the average of the two Em and Ee values so obtained being used to correct the measured CH 4 concentrations. For onboard measurement, the efficiency of the NMC may be assessed in a laboratory before and after the measurement exercise.

With the agreement of the Administration, alternative approaches to the assessment of NMC efficiency may be accepted.

Methane efficiency < 15%

Ethane efficiency > 98%

8.5.1 NMC methane efficiency

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exercise.

With the agreement of the Administration, alternative approaches